The Redistribution Operator: Non-Associativity as the Splitting Mechanism, with Zero Divisors as Saturation

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A companion to the sedenion split classification, deriving the exact spectral structure underlying Becoming §4's Moreno/Reggiani citations, and identifying its mechanism as the associator itself.
Status: Revision 4. Restructured for the working corpus: revision-line process narrative trimmed to a bare statement.

Abstract. For $A = v_1 + v_2 e_8 \in \mathbb{S}$, the coupling block in $M(A)^\top M(A)$ — the off-diagonal term whose structure determines the spectrum's departure from uniformity — is exactly $B = [L_{v_1}, R_{v_2}] \circ \tau$, the commutator of left- and right-multiplication composed with conjugation; since scalar multiples of the identity commute with everything, this reduces immediately to $B = [L_u, R_w] \circ \tau$ for the imaginary parts $u, w \in \mathrm{Im}(\mathbb{O})$ alone, and the commutator is exactly the associator operator: $[L_u, R_w](y) = -(u, y, w)$. Verified to machine precision. Three consequences follow, each now proven rather than sampled or conjectured. *First*, the scalar parts of v_1, v_2 cancel identically in B, so the spectral formula $\mathrm{spec}(L_A^\top L_A) = \{N(A) - \beta\ (\times 4),\ N(A)\ (\times 8),\ N(A) + \beta\ (\times 4)\}$, $\beta = \sqrt{D_1(A)^2 - D_2(A)}$ in Koebisu's own factors (arXiv:2512.13002), holds on all of $\mathbb{S}$, not merely the imaginary case Revision 2 derived. *Second*, $\ker B = \mathrm{alg}(u, w) = \mathrm{span}\{1, u, w, uw\}$ exactly when $u \wedge w \neq 0$ — the quaternion subalgebra generated by u, w — because Artin's theorem forces any two-generated subalgebra of an alternative algebra to be associative, and the associator kills exactly what associates (unconditionally; equality with a 4-dimensional kernel needs the nondegenerate case). This is why $\dim \ker B = 4$ identically off the degenerate locus: it is the dimension of a quaternion algebra, not an empirical regularity. *Third*, on the complement of this subalgebra the coupling acts as exactly β times an isometry, derived from an explicit associator identity in the doubling $\mathbb{O} = \mathbb{H} \oplus \mathbb{H}e_4$ — proven here after an initial proof attempt failed on a missing orthogonality hypothesis, corrected and reported alongside the final result. The mechanism is now named precisely: **the middle band is the associative directions, where the coupling vanishes by Artin; the outer bands are the anti-associative complement; zero divisors are exactly where the associator coupling saturates the composition norm.** The physical question this structure opens for fermionic construction is unchanged and remains open.

1. Background, corrected: what Theorems 1–2 forbid, precisely

Formal Beginning §4 proves two theorems about $\mathbb{S}$'s own multiplication. **[THM 1]** No nonzero real $A \in \mathbb{S}$ satisfies $A^2 = 0$. This holds for *all* of $\mathbb{S}$, not only the imaginary case: writing $A = \alpha + v$ ($\alpha \in \mathbb{R}$, v imaginary), $A^2 = (\alpha^2 - N(v)) + 2\alpha v = 0$ forces $\alpha v = 0$ and $\alpha^2 = N(v)$; if $v \neq 0$ then $\alpha = 0$ and $N(v) = 0$, forcing $v = 0$ too — either way $A = 0$. **[THM 2]** No nonzero real A satisfies $L_A^2 = 0$ as an operator: $L_A^2(e_0) = A \cdot A = A^2$, so $L_A^2 = 0$ would force $A^2 = 0$, reducing to Theorem 1 directly, for any A.

[FACT, corrected] Revision 2 introduced a genuine internal contradiction here: Theorem 1 was stated with an erroneous "for A purely imaginary" qualifier, while the very next paragraph correctly invoked Theorem 1 "for any A, imaginary or not." The qualifier belonged to a separate, narrower fact — the trace identity $\operatorname{tr}(L_A^2) = -16N(A)$, which does hold only for imaginary A (general $A = \alpha + v$ gives $\operatorname{tr}(L_A^2) = 16(\alpha^2 - N(v))$, confirmed exactly across 10 general trials) — and that qualifier was misattached to the theorem instead. Corrected here: Theorem 1 is unrestricted; the trace identity alone carries the imaginary-only scope, and — as already noted — is not load-bearing for Theorem 2's proof regardless.

[FACT, verified] For unit-norm $A \in \mathbb{O} \cdot e_8$ specifically, $L_A^2 = -1$ exactly — all sixteen eigenvalues identically -1, unaffected by the above.

2. Zero divisors: what is, and is not, distinctive

[DEF] A zero divisor of $\mathbb{S}$ is a nonzero A admitting some nonzero $B \neq A$ with $AB = 0$. Becoming §4 cites: normalized zero-divisor *pairs* isometric to G_2 (Moreno, 1998); the zero-divisor *locus* isometric to $V_2(\mathbb{R}^7)$, homogeneous under G_2 (Reggiani, arXiv:2411.18881); the explicit combinatorics due to de Marrais (arXiv:math.GM/0011260).

[FACT] The trace identity $\operatorname{tr}(L_A^\top L_A) = 16N(A)$ holds for *every* $A \in \mathbb{S}$ — every column of L_A has norm exactly $|A|$, a fact independent of zero-divisor structure. This is not distinguishing; §3 identifies what actually is.

3. The mechanism: the coupling block is the associator

[SETUP] Write $A = v_1 + v_2 e_8$. The left-multiplication matrix, in block form (Koebisu, arXiv:2512.13002, Lemma 3.4, cross-validated against this project's own independently-built multiplication table to zero defect), is

$$M(A) = \begin{pmatrix} L_{v_1} & -R_{v_2} \circ \tau \\ L_{v_2} \circ \tau & R_{v_1} \end{pmatrix}.$$

[THM, proven, the identification — general from the start] For $A = v_1 + v_2 e_8$ with $v_1, v_2 \in \mathbb{O}$ arbitrary (no restriction to imaginary parts), the $(1,2)$ block of $M(A)^\top M(A)$ is

$$L_{v_1}^\top(-R_{v_2} \circ \tau) + (L_{v_2} \circ \tau)^\top R_{v_1} = -L_{\bar{v}_1} R_{v_2} \circ \tau + \tau \circ L_{\bar{v}_2} R_{v_1}$$

using the composition-algebra adjoint identities $L_x^\top = L_{\bar{x}}$, $R_x^\top = R_{\bar{x}}$ (obtained by polarizing the composition law $N(xy) = N(x)N(y)$ into $\langle xy, z\rangle = \langle y, \bar{x}z\rangle$; standard, e.g. Schafer, cited below — confirmed here exactly for general $x \in \mathbb{O}$). Inserting $\tau\tau = 1$ twice and using $\tau L_{\bar{v}_2} \tau = R_{v_2}, \tau R_{v_1} \tau = L_{\bar{v}_1}$ (the conjugation-swap identities, likewise verified for general elements, not only imaginary ones) gives

$$-L_{\bar{v}_1} R_{v_2} \circ \tau + R_{v_2} L_{\bar{v}_1} \circ \tau = -[L_{\bar{v}_1}, R_{v_2}] \circ \tau.$$

Writing $v_1 = x_1 + u, \bar{v}_1 = x_1 - u$: the scalar part $x_1 \cdot 1$ drops from any commutator, so $[L_{\bar{v}_1}, R_{v_2}] = -[L_u, R_{v_2}]$, and by the same argument on $v_2 = x_2 + w, = -[L_u, R_w]$. Hence

$$B = -[L_{\bar{v}_1}, R_{v_2}] \circ \tau = [L_u, R_w] \circ \tau, \qquad [L_u, R_w](y) = u(yw) - (uy)w = -(u, y, w),$$

general from the first line: no separate "imaginary case, then extend" step is needed. The coupling block *is* the associator operator, composed with conjugation. The full three-step identity and its

intermediate form are each verified exactly ($\sim 10^{-15}$ on general elements, $\sim 5\times 10^{-15}$ on the intermediate commutator form). *B is skew:* the commutator $[L_u, R_w]$ of two skew operators ($L_u^\top = -L_u$, $R_w^\top = -R_w$ for imaginary u, w) is itself skew — but $B = [L_u, R_w] \circ \tau$ needs the further fact that τ commutes with $[L_u, R_w]$, since τ alone is symmetric, not skew. This follows from the same swap-antisymmetry stated above: $\tau L_u \tau = -R_u$ and $\tau R_w \tau = -L_w$, so conjugating the commutator gives $\tau[L_u, R_w]\tau = [R_u, L_w]$; and $[R_u, L_w] = [L_u, R_w]$ by the same alternation identity used for the swap. Together, $\tau[L_u, R_w]\tau = [L_u, R_w]$, which is exactly what makes B skew. B also satisfies $B(w, u) = -B(u, w)$, writing $B(u, w)$ for the coupling block as a function of the ordered pair (swapping u, w flips the associator's sign) — confirmed directly, and used again below.
[LEMMA, proven, the diagonal blocks — general form] $L_{\bar{v}} L_v = N(v)\cdot 1$ for *any* octonion v, imaginary or not: from the composition identity $\langle vy, vz\rangle = N(v)\langle y, z\rangle$, so both diagonal blocks of $M(A)^\top M(A)$ equal $N(A)\cdot 1$ directly, for general v_1, v_2 — no separate extension step required, and confirmed to machine precision.
[ASSEMBLY, proven] With both diagonal blocks equal to $a1 := N(A)1$ and off-diagonal blocks $\pm B$, B skew with $\operatorname{rank}(B) = 2k$: on each of B's k two-dimensional rotation planes (singular value β, real skew-symmetric matrices having singular values in equal pairs per plane), the 4×4 real block restricted to that plane-pair — two real dimensions from each copy — diagonalizes with eigenvalues $a+\beta$ and $a-\beta$, each with multiplicity 2; the trivial directions ($\ker B$, dimension $8-2k$ in each copy) contribute eigenvalue a with multiplicity $2(8-2k)$. Consequently, once $\dim\ker B = 4$ and a single distinct nonzero singular value β are established below, the spectral formula

$$\operatorname{spec}(L_A^\top L_A) = \{N(A)-\beta\ (\times 4),\ N(A)\ (\times 8),\ N(A)+\beta\ (\times 4)\}, \qquad \beta = \sqrt{D_1(A)^2 - D_2(A)}$$

holds on **all of** $\mathbb{S}$, general from the first line of the derivation above — no restriction to the imaginary case anywhere in the chain.
[THM, proven — Revision 2's first open item, closed] $\operatorname{alg}(u, w) = \operatorname{span}\{1, u, w, uw\} \subseteq \ker B$ unconditionally: $\operatorname{alg}(u, w)$, being generated by two elements in an alternative algebra, is associative by Artin's theorem, and the associator vanishes identically on associative triples — so B, built from the associator, annihilates it regardless of u, w. *Equality* $\ker B = \operatorname{alg}(u, w)$ — hence $\dim\ker B = 4$ exactly — holds when $u\wedge w \neq 0$ (that is, u, w linearly independent): containment already fixes $\dim\ker B \geq 4$, and equality follows because the complement isometry established next is then nonzero, leaving no room for a larger kernel. (When $u\wedge w = 0$, $\operatorname{alg}(u, w)$ collapses to a lower-dimensional associative span and $\ker B$ is correspondingly larger — the degenerate case identified in the Corollary below.) Verified across 5 trials: B annihilates the predicted 4-dimensional basis to machine precision, with $\dim\ker B = 4$ exactly whenever $u\wedge w\neq 0$.
[THM, proven — Revision 2's second open item, closed, with a reported self-correction] On the orthogonal complement of $\operatorname{alg}(u, w)$ (nonempty when $u\wedge w\neq 0$), B acts as exactly $\beta = 2\sqrt{\|u\|^2\|w\|^2 - \langle u, w\rangle^2}$ times an isometry. *Setup:* by alternativity $(u, y, u) = 0$, so only the component of w orthogonal to u — call it $w_\perp$ — contributes; write a, b for an orthonormal pair spanning $\operatorname{span}\{u, w_\perp\}$, so $a\perp b \in \operatorname{Im}(\mathbb{H})$ for the quaternion subalgebra $\mathbb{H} = \operatorname{alg}(a, b)$, and any octonion decomposes as $\mathbb{O} = \mathbb{H}\oplus\mathbb{H}\ell$ for a unit $\ell\perp\mathbb{H}$ (the Cayley–Dickson decomposition induced by any quaternion subalgebra — the same fact underlying Keystone's own closed-subspace structure — which is what replaces a G_2-normal-form reduction here: it applies directly to *any* (u, w), aligned or not, with no reduction step needed). One further fact used silently by both this theorem and the

kernel theorem above, worth stating once, in its genuinely general form: τ preserves $\mathrm{alg}(u,w) = \mathrm{span}\{1,u,w,uw\}$ setwise, for *all* u,w, unconditionally — the fact the kernel theorem's general (not-necessarily-orthogonal) case actually needs. This follows from one line: conjugation is reflection through the real axis, $\bar{x} = 2\langle x,1\rangle \cdot 1 - x$ (verified exactly), a linear combination of 1 and x alone — so τ maps *any* subspace containing 1 into itself, $\mathrm{alg}(u,w)$ included, with no orthogonality needed anywhere. What orthogonality buys, specifically for the orthonormal pair a,b used in this theorem, is a simplification of that general fact's action on the product term: $\tau(ab) = ba = -ab - 2\langle a,b\rangle$ (the general anticommutation identity $xy + yx = -2\langle x,y\rangle$, confirmed exactly) collapses to the clean negation $\tau(ab) = -ab$ precisely when $\langle a,b\rangle = 0$ — the case in force here, which is what lets kernel and isometry statements transfer cleanly between $[L_u,R_w]$ and $B = [L_u,R_w]\circ\tau$ in this theorem's own orthonormal setup. *The identity, proven:* for $h \in \mathbb{H}$ and $\ell \in \mathbb{H}^\perp$, the doubling formula on $\mathbb{H} \oplus \mathbb{H}\ell$ gives $a(h\ell) = (ha)\ell$ and $(h\ell)b = (h\bar{b})\ell$ (direct expansion of the Cayley–Dickson product), so

$$(a,h\ell,b) = \big(a(h\ell)\big)b - a\big((h\ell)b\big) = \big((ha)\bar{b}\big)\ell - \big((h\bar{b})a\big)\ell = \big((ha)\bar{b} - (h\bar{b})a\big)\ell.$$

For a,b imaginary, $\bar{a} = -a, \bar{b} = -b$, and associativity of $\mathbb{H}$ gives $(ha)\bar{b} = -h(ab)$, $(h\bar{b})a = -h(ba)$, so the bracket is $h(ba) - h(ab) = h(ba - ab)$; for $a \perp b$, $ab = -ba$ (the quaternion product of orthogonal imaginaries anticommutes exactly), so $ba - ab = -2ab$, giving

$$(a,\ h\ell,\ b) = -2\,\big(h(ab)\big)\,\ell \qquad (a \perp b \in \mathrm{Im}(\mathbb{H}),\ h \in \mathbb{H},\ \ell \in \mathbb{H}^\perp),$$

verified to $\sim 2\times 10^{-14}$ across 10 trials on top of the symbolic proof (an earlier verification attempt that placed ℓ inside $\mathbb{H}$ itself trivially found zero on both sides, since any associator computed entirely within an associative subalgebra vanishes — a construction error in the check, caught and corrected before use, not a property of the identity; a separate, genuine error was caught earlier still — an initial proof draft omitted the $a \perp b$ hypothesis entirely and failed numerically, defect ~ 39, before this corrected version was found). *Closing the isometry claim:* the map $h \mapsto h(ab)$ is right-multiplication by the fixed quaternion ab, hence exactly $|ab| = |a||b|$ times an isometry of $\mathbb{H}$ (quaternion right-multiplication is norm-scaled-isometric because the quaternion norm is multiplicative) — giving all four singular values of $h \mapsto -2h(ab)$ identically equal to $2|a||b|$, and, restoring the original normalization $a = u/\|u\|,\ b = w_\perp/\|w_\perp\|$, $\beta = 2\|u\|\|w_\perp\| = 2\sqrt{\|u\|^2\|w\|^2 - \langle u,w\rangle^2}$ uniformly. No G_2-reduction is needed: Artin's theorem, the induced Cayley–Dickson decomposition, and one quaternionic computation are the whole mechanism.

[COR, the unified picture] One expression accounts for every case:

Zero divisors ($D_2 = 0$: $\|u\| = \|w\|, \langle u,w\rangle = 0$): $\beta = D_1$ exactly, the lower band saturates to zero — the genuine kernel. For $A = e_1 + e_{10}$: spectrum $\{0^4, 2^8, 4^4\}$.

Generic elements ($D_2 > 0$): $0 < \beta < D_1$, three genuinely distinct eigenvalues, same $4/8/4$ multiplicities.

$u \wedge w = 0$ (parallel or one vanishing — strictly broader than "$u = 0$," confirmed via $A = e_1 + e_9$, $u = w = e_1$, giving the exactly uniform spectrum $2 \cdot 1$): $\beta = 0$, all three bands collapse — the geometric condition is coplanarity of u,w, not merely one vanishing.

[COR, proven — the sharp norm bound, with its equality case] $\|xy\| \le \sqrt{2}\,\|x\|\|y\|$ on all of $\mathbb{S}$, and the bound is attained exactly, not merely approached. *Proof:* $\|L_A\|^2 = N(A) + \beta$ (the top eigenvalue of $L_A^\top L_A$, above), and

$$\beta = 2\sqrt{\|u\|^2\|w\|^2 - \langle u, w\rangle^2} \overset{(1)}{\leq} 2\|u\|\|w\| \overset{(2)}{\leq} \|u\|^2 + \|w\|^2 \overset{(3)}{\leq} N(A),$$

giving $\|L_A\|^2 \leq 2N(A)$, i.e. $\|Ay\| \leq \sqrt{2}\,\|A\|\|y\|$ for all y — already the fully general bound, since A ranges over all of $\mathbb{S}$. Each inequality is tight under exactly one of Moreno's three defining clauses of the zero-divisor condition — not a coincidence, but the same chain read three ways: (1) is Cauchy–Schwarz, equality iff $\langle u, w\rangle = 0$; (2) is AM–GM, equality iff $\|u\| = \|w\|$; (3) is equality iff $x_1 = x_2 = 0$ (no scalar content diluting the norm). All three hold simultaneously exactly on the zero-divisor locus, where $\beta = D_1 = N(A)$ and the bound saturates completely — this chain, run as displayed, is precisely the statement $\beta \leq N(A)$, i.e. the positive-semidefiniteness of the lower band ($N(A) - \beta \geq 0$ always). Verified exactly: for $A = e_1 + e_{10}$, the top eigenvector of $L_A^\top L_A$ gives $\|Av\|/(\|A\|\|v\|) = 1.414213562373095$ against $\sqrt{2} = 1.414213562373095$, matching to the last displayed digit. (A direct 3000-sample random search over $\mathbb{S} \times \mathbb{S}$ finds only ≈ 1.27 — unsurprising and not evidence against exact attainment, since the maximizing pairs form a measure-zero locus that random sampling essentially never hits; the analytic argument above, not sampling, is what establishes the bound.) This completes §4's saturation claim below: "$\beta = D_1$ is the maximum the coupling can reach" has its proof here. The bound's literature status is left open deliberately: it follows naturally enough from the algebra's own structure that it likely already exists in the composition-algebra literature (Moreno, Biss–Dugger–Isaksen, or the general Cayley–Dickson norm-defect literature), and no claim of novelty is made here without checking first.

4. Interpretation

The splitting mechanism is non-associativity itself. The middle band ($N(A)$, multiplicity 8) is exactly the associative directions — $\mathrm{alg}(u, w)$ paired across both copies of $\mathbb{O}$ — where the coupling vanishes by Artin's theorem, a structural fact requiring no computation. The outer bands are the anti-associative complement, where the associator acts as a genuine isometric coupling of strength β. Zero divisors are exactly the locus where this coupling *saturates* the composition norm: $\beta = D_1$ is the maximum the associator coupling can reach relative to $N(A)$ — proven above, closing §3, via the same three-inequality chain that gives the sharp norm bound — and saturation is precisely the zero-divisor condition. This is a structural account, not a numerical description: the redistribution is non-associativity made spectrally visible, with an exact saturation threshold rather than an observed pattern.

5. Scope

Three items, stated precisely.

(i) This note does not claim, and does not attempt to construct, a working fermionic creation operator. $\theta^2 = 0$ is not satisfied — Theorem 1 forbids it — and the kernel of L_A for a zero divisor does not contain A itself.

(ii) **The open physical question, unchanged (registry item REG-6, shared with Becoming §4):** does fermionic construction require exact operator-nilpotency, or only an annihilation structure with the right statistics — and can this structure supply the latter? The bosonic sector's own Weyl algebra arose by contraction, with native content living in what the limit discards; whether the same shape holds here is unknown. The counterweight stands with equal force: Clifford algebras are rigid, and the discipline is not to reach for deformed relations without demonstrated need. Nobody knows the answer; it remains open, not resolved.

(iii) **The box-kite matching question, narrowed but not closed.** De Marrais's catalog classifies the

zero-divisor locus combinatorially. The bridge object is now named exactly: $\mathrm{alg}(u, w)$, the quaternion subalgebra each zero divisor generates, is precisely the kind of data the assessor classification organizes around. Whether the correspondence is exact remains unchecked.

6. Provenance

This computation originated from a conversational question about "almost nilpotent" structure in $\mathbb{O} \cdot e_8$. Revision 1 found the three-band spectrum computationally, verified on nine sampled cases. Revision 2 derived it in closed form via Koebisu's block decomposition. Revision 3 identified the coupling block exactly with the associator, closing two previously-conjectured facts. Revision 3.1 restored two derivation steps a restructure had dropped and inverted an inside-out epistemic framing on the norm bound. This revision, 3.2, closes the one remaining gap: the identification theorem itself — that the coupling block equals $[L_u, R_w] \circ \tau$ at all — had been carried by sampling alone under a "proven" tag since Revision 3; its three-line derivation is now displayed, general from the first step, with no [THM, proven] tag in the document resting on numerical verification alone. Placement of the norm-bound corollary and its citing pointers are corrected to match. The reconvergence pattern recorded before continues; the note is, as far as this process can currently establish, complete.

References

Companion document: *The Formal Beginning* (Theorems 1–2, §4).

Companion document: *Becoming*, §4 (Moreno/Reggiani/de Marrais citation stack).

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de Marrais, R. P. C., "The 42 Assessors and the Box-Kites They Fly," arXiv:math.GM/0011260 (2000).

Koebisu, S., "Determinant Factorization for Left Multiplication in the Sedenions," arXiv:2512.13002 (2025).

Artin's theorem on two-generated subalgebras of alternative algebras: see Schafer, R. D., *An Introduction to Nonassociative Algebras*, Academic Press (1966).

Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author's responsibility.